

Structural and magnetic properties of $\text{Mn}_{1+b}\text{Fe}_{2-b}\text{O}_{4-\delta}$ NPs:

A view from a combination of EDS and XAS



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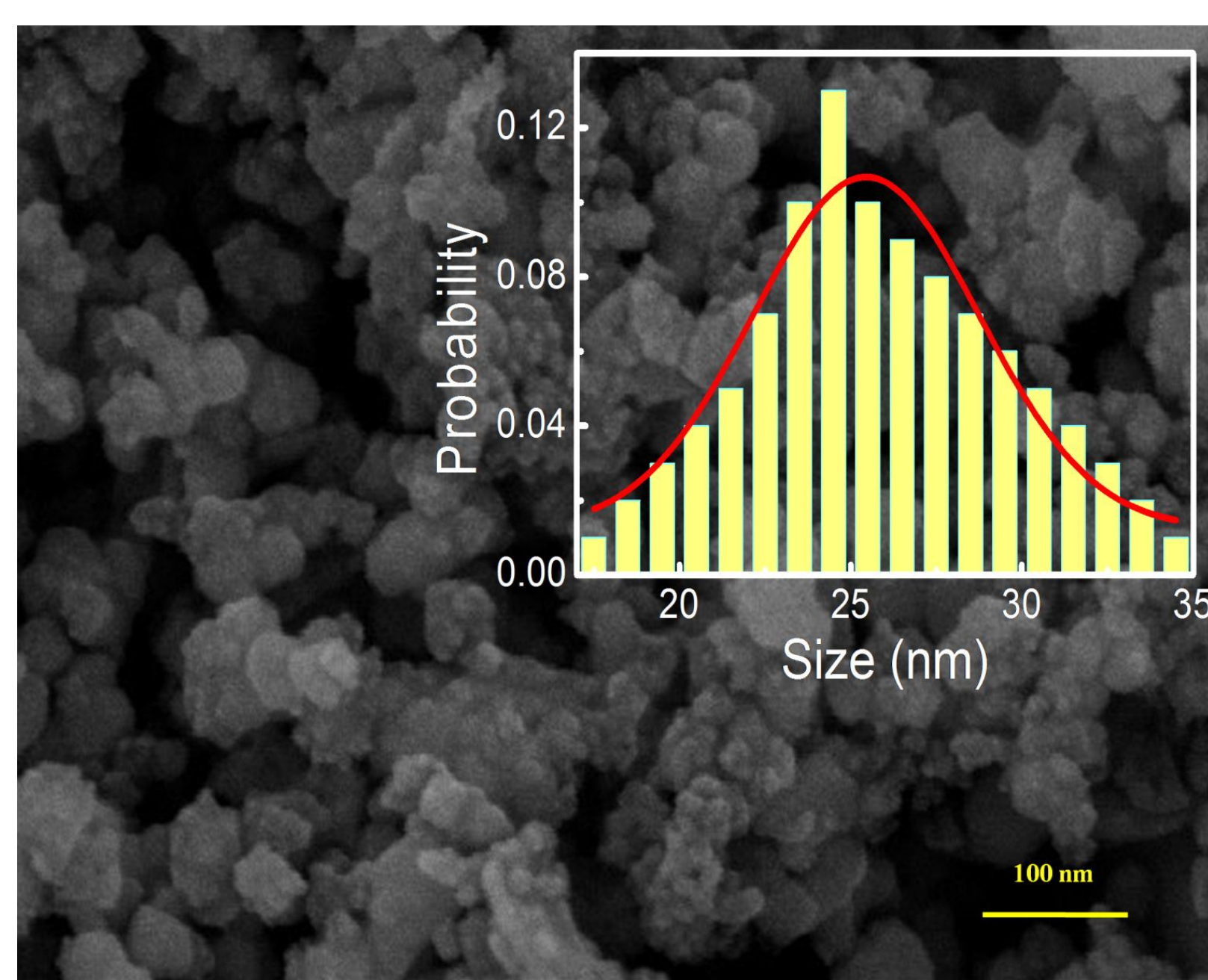
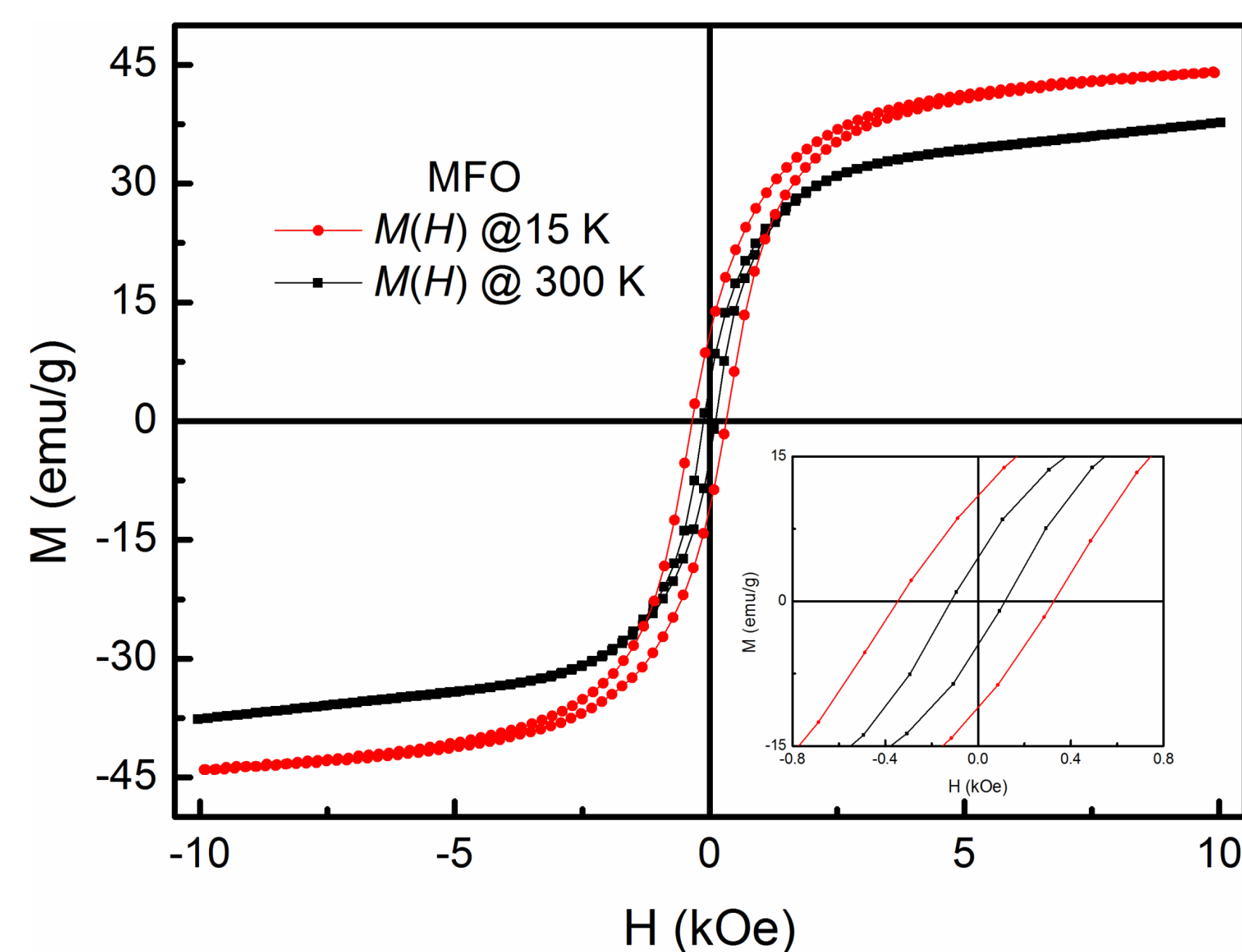
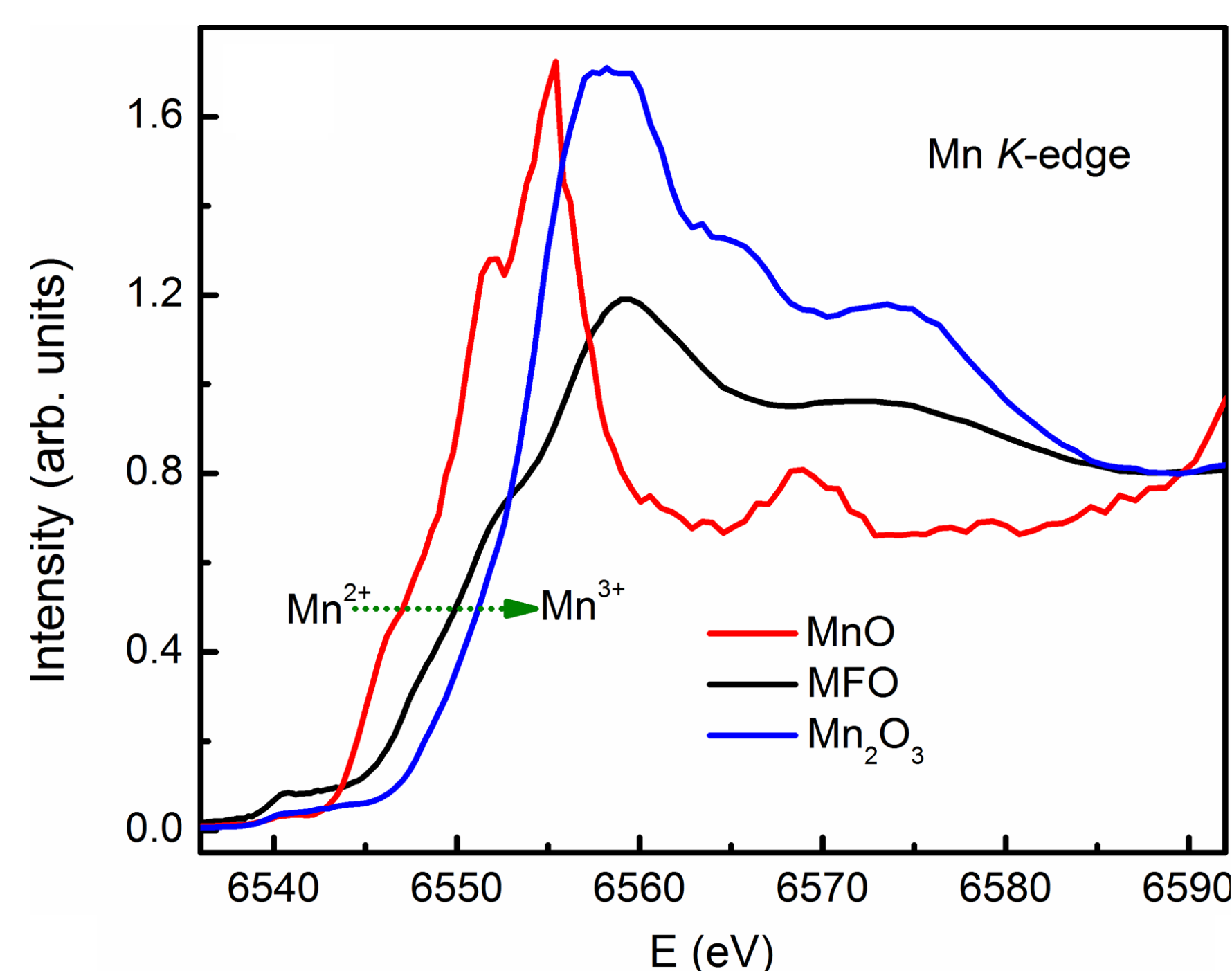
Introduction

- ❖ Spinel oxides with the general chemical formula of $(A)[B_2]O_4$ are a class of ferrimagnetic compounds, in which () and [] represent the tetrahedral (tet.) and octahedral (oct.) sites.
- ❖ Manganese ferrite (MnFe_2O_4) is an inverse spinel, where two magnetic sub-lattices are antiferromagnetically aligned due to $(\text{Fe}^{3+})[\text{Mn}^{2+}\text{Fe}^{3+}]O_4/(\text{d}^5)[\text{d}^5\text{d}^5]$.
- ❖ Under certain synthesis conditions to obtain samples in the nanoscale, the distribution of Fe and Mn ions among (A) and [B] sites is altered.
- ❖ Many studies have examined the cation distribution in MnFe_2O_4 NPs. However, those studies only used a single tool (XRD or Mossbauer spectra) or a combination of XRD and XANES tools.
- ❖ In this work, we have prepared and studied magnetic properties of $\text{Mn}_{1+b}\text{Fe}_{2-b}\text{O}_{4-\delta}$ (MFO) NPs by combining XRD, EDS and XAS. Other factors affected the magnetic properties were also studied in detail.

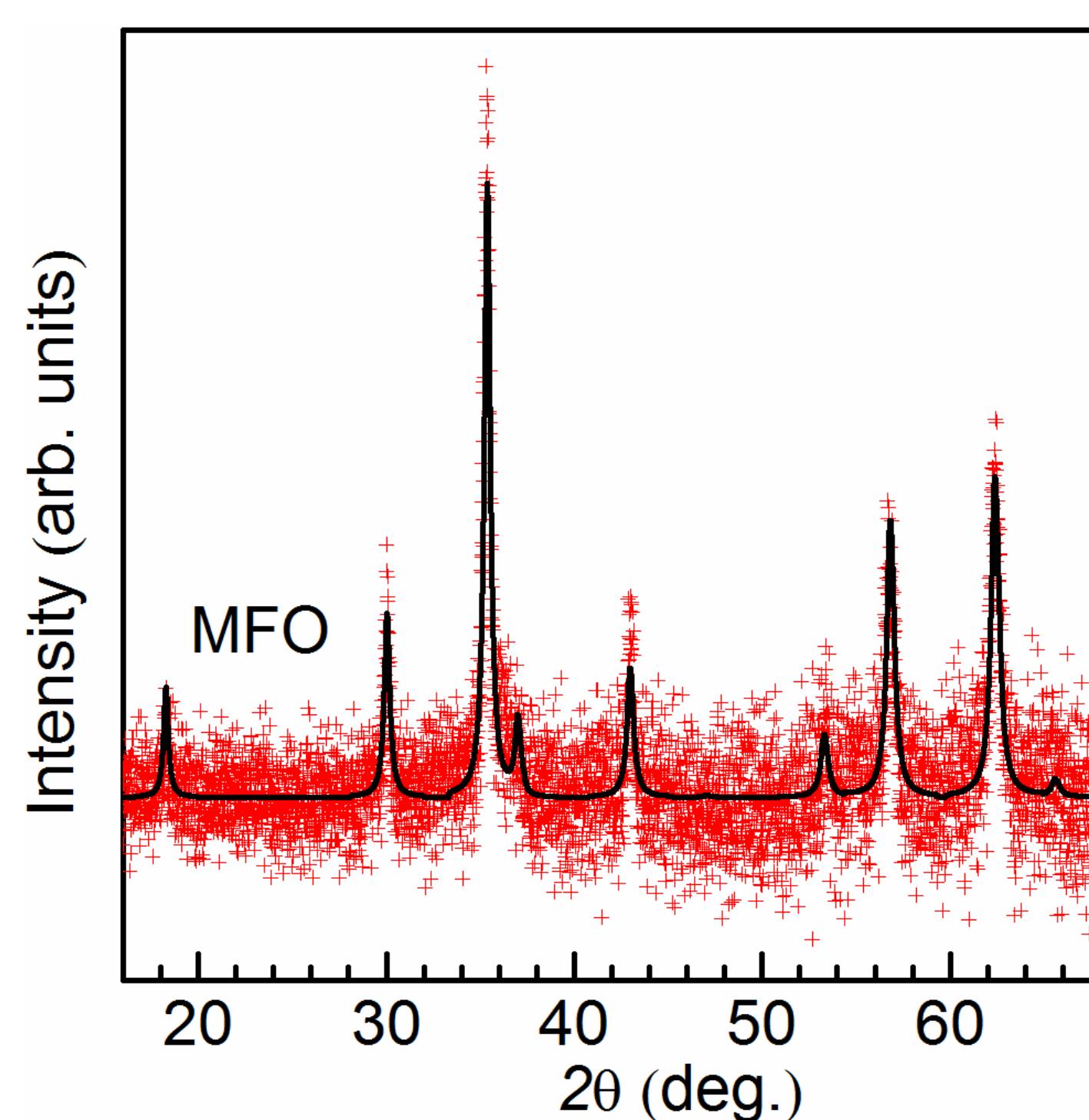
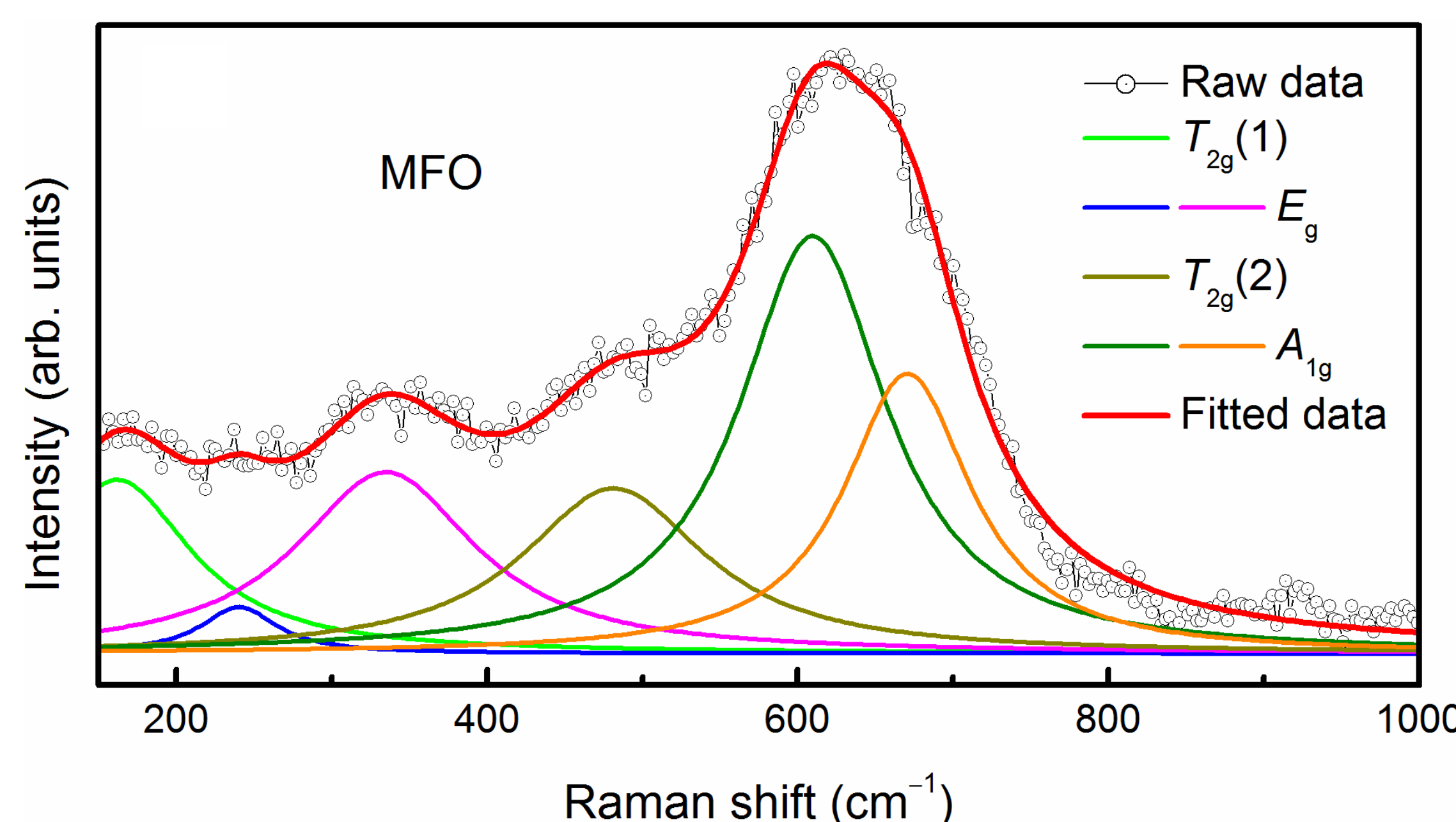
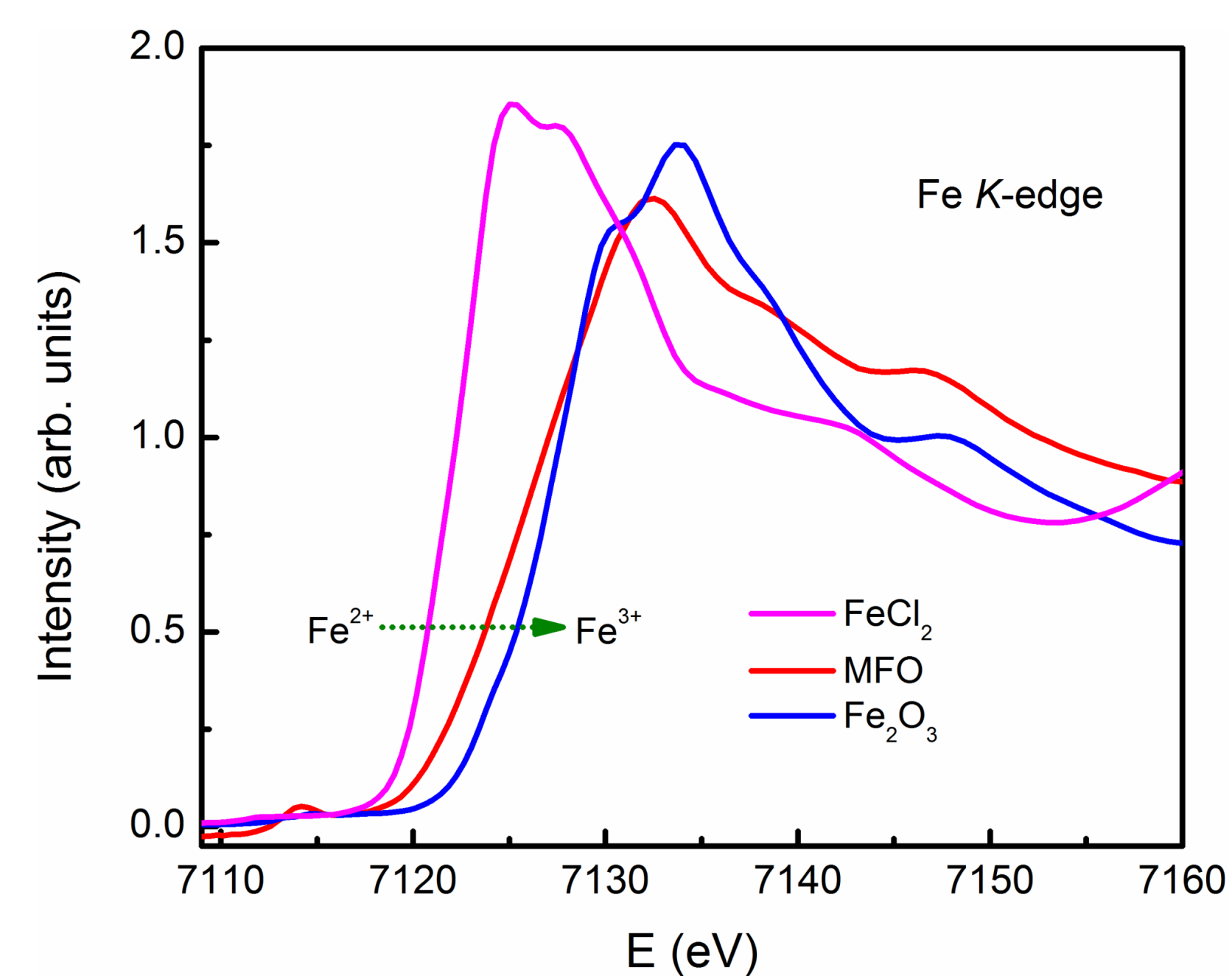
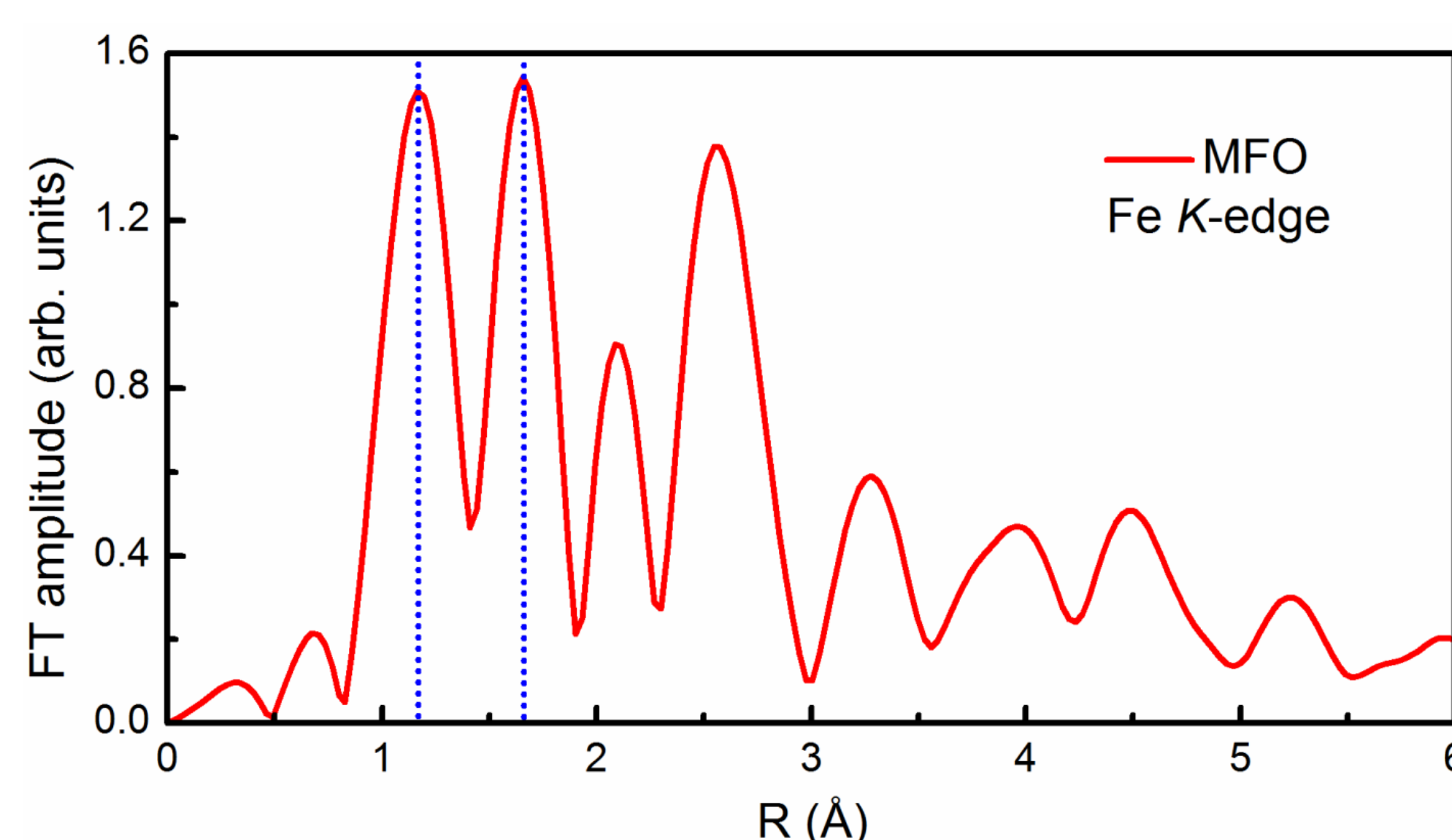
Experiment

- ❖ $\text{Mn}_{1+b}\text{Fe}_{2-b}\text{O}_{4-\delta}$ (MFO) NPs were prepared by hydrothermal method.
- ❖ Surface morphology and chemical composition of MFO NPs were analyzed using FE-SEM equipped with EDS.
- ❖ Crystal structure was checked by using XRD and Raman scattering (RS) spectroscopy.
- ❖ Electronic structures were studied using X-ray absorption spectroscopy (XAS).
- ❖ Magnetic measurements were carried out using a VSM.

Results & Discussion

**Fig. 1.** SEM image of MFO NPs. Inset: Histogram showing particle size distribution of the sample.**Fig. 3.** $M(H)$ loops of MFO NPs recorded at 15K and 300K.**Fig. 6.** XAS spectra of MFO NPs recorded at the Mn K-edge.**Table 3.** Estimations of the cation distribution for MFO NPs using EDS and XAS.

Sample	A site		B site			n _{cal}	n _{exp_300}	n _{exp_15}
	Mn ²⁺	Fe ³⁺	Mn ³⁺	Fe ³⁺	Fe ²⁺	(μ _B /f.u.)	(μ _B /f.u.)	(μ _B /f.u.)
MFO	0.07	0.93	1.05	0.29	0.66	3.29	1.13	1.66

**Fig. 2.** XRD pattern of MFO NPs refined by the Rietveld method.**Fig. 4.** RS spectra of MFO NPs excited by wavelength $\lambda_{\text{exc}} = 532$ nm.**Fig. 5.** XAS spectra of MFO NPs recorded at the Fe K-edge.**Fig. 7.** k^3 -weighted FT XAFS spectra at the Fe K-edge for MFO NPs.

Summary

- ✓ We used a hydrothermal method to synthesize cubic MFO NPs.
- ✓ RS confirmed that MFO NPs belong to the mixed spinel group.
- ✓ The extraordinary magnetic properties have been clarified in detail by combining EDS, XAS, and VSM data.
- ✓ Other factors (spin canting, thermal stability, and magnetically dead layer) affected the magnetic properties were discussed in detail.

- The atomic normalization of O in MFO NPs (3.74) can be negotiable according to the error of EDS measurement.
- Doublet-like peak A_{1g} confirmed that MFO NPs is mixed spinel.
- According to **Table 2**, **Fig. 5**, **Fig. 6**, and **Fig. 7**, one derived to cation distribution of MFO NPs, shown in **Table 3**.
- The difference in magnetization between calculation n_{cal} and experiment n_{exp} can be attributed to spin canting ($\alpha_{\text{YK}} \approx 36.5^\circ$) and magnetically dead layer ($t = 3.5$ nm).